8. Pythagoras's Theorem and its extensions

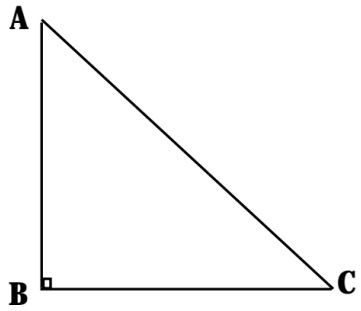
Theorem of Pythagoras

In a right angled triangle the square described on the hypotenuse is equal to the sum of the squares described on the other two sides.

If ABC is a right angled triangle and

$\angle b = 90^0$,

Then $AC^2 = AB^2 + BC^2$



Converse of the above theorem

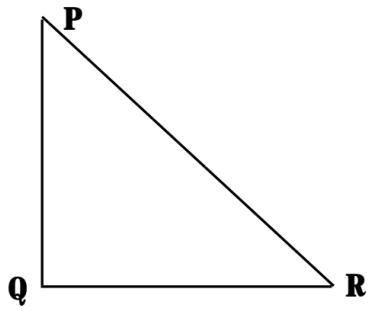
Theorem

If the square described on one side of the triangle equal to the sum of the squares described on the other two sides, then the angle contained by these two sides is a right angle.

In the triangle PQR if

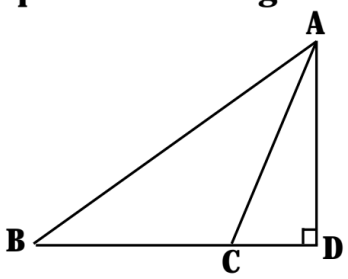
$PQ^2 + QR^2 = PR^2$

then $\angle PQR = 90^0$



Theorem

In an obtuse angled triangle the square on the side opposite to the obtuse angle is equal to the sum of the squares on the sides containing the obtuse angle plus twice the rectangle contained by either of these sides and the projection on it of the other.



Given : **ABC is a triangle with an obtuse angle at C. CD is the projection of AC upon BC.**

To Prove : $AB^2 = BC^2 + CA^2 + 2CB \cdot CD$

Proof : **ABD is a right angled triangle.**

$AB^2 = BD^2 + AD^2$ (Pythagoras's theorem)

$= (BC + CD)^2 + AD^2$

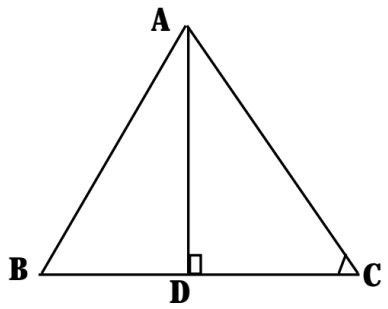
$= BC^2 + 2\ BC \cdot CD + CD^2 + AD^2$

$= BC^2 + 2\ BC \cdot CD + AC^2$

$= BC^2 + AC^2 + 2\ BC \cdot CD$

Theorem:

In any triangle the square on the side opposite to an acute angle is equal to the sum of the squares on the sides containing the acute angle less twice the rectangle contained by one of those sides and the projection on it of the other.



Given : **ABC is a triangle with an acute angle at C. CD is the projection of AC upon BC.**

To prove : $AB^2 = CA^2 + CB^2 - 2\ CB \cdot CD$

Proof : ABD is a right angled triangle.

$AB^2 = AD^2 + BD^2$

$= AD^2 + (BC - CD)^2$

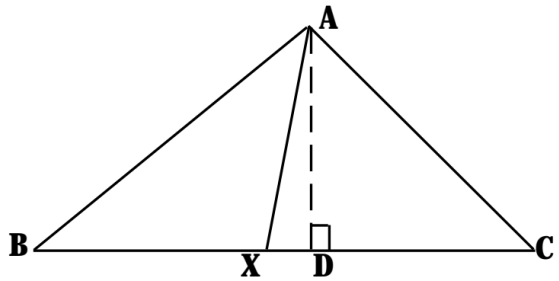
$= AD^2 + BC^2 - 2\ BC \cdot CD + CD^2$

$= AD^2 + CD^2 + BC^2 - 2\ BC \cdot CD$

$= AC^2 + BC^2 - 2\ BC \cdot CD$

Apollonius' Theorem:

The sum of the squares on two sides of a triangle is equal to twice the sum of the squares on half the third side and twice the square on the median which bisects that side.



Given : **ABC is a triangle. BX = XC. AX is a median.**

To prove : $AB^2 + AC^2 = 2BX^2 + 2AX^2$

Construction : Draw AD perpendicular to BC.

Proof : Of the angles AXB, AXC, one is obtuse and the other is acute.

Let the angle AXB is obtuse.

In triangle AXB,

$AB^2 = AX^2 + BX^2 + 2\,AX\,.\,XD$ (1)

In triangle AXC,

$AC^2 = AX^2 + XC^2 - 2\,AX\,.\,XD$ (2)

Adding (1) and (2),

$AB^2 + AC^2 = 2\,AX^2 + 2\,BX^2$ (since BX = XC)

Excercise 8

1 **ABC is an equilateral triangle. O is the mid point of BC. Prove that $3\,BC^2 = 4\,OA^2$.**

2. ABCD is a square of side 12 cm length. Prove that the area of the square described on the diagonal BD is twice the area of the square ABCD.

3. PQR is a triangle right angled at Q. Mid points of QR and PQ are X and Y respectively. Show that $6PR^2 = 4(PX^2 + RY^2)$

4. If from any point O within a triangle ABC, perpendiculars OX, OY, OZ are drawn to BC, CA, AB respectively. Show that $AZ^2 + BX^2 + CY^2 = AY^2 + CX^2 + BZ^2$

5. In a triangle ABC, AD is drawn to perpendicular to BC. Let p denote the length of AD.

   (i) If a = 25 cm, p = 12 cm, BD = 9 cm find b, c

   (ii) If b = 82 cm, c = 1 cm, BD = 60 cm; find p and c and prove that $\sqrt{b^2 - p^2} + \sqrt{c^2 - p^2} = a$ (AB = c, BC = a, CA = b)

6. ABC is a triangle right angled at C, and p is the length of the perpendicular from C on AB. By expressing the area of the triangle in two ways, show that pc = ab.

   Hence deduce $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$ (BC = a, CA = b, AB = c)

7. In triangle ABC, AD is a median. Point X is on the side BC, so that BX = XD and $A\hat{X}B = 90^0$. Prove that $4(AC^2 - AD^2) = BX^2$

8. ABD is a triangle right angled at A. Point C is in the side BD so that 2 BC = CD and $A\hat{C}D = 90^0$. If CT is a median of the triangle ACD prove that $2(CT^2 + AT^2) = AD^2$

9. In a triangle ABC, the points E and D are taken on BC such that BE = ED = DC. Prove that $AB^2 + AE^2 = AC^2 + AD^2$

10. In a ΔABC, D is the mid point of BC. Find the length of the median AD when AB = 4 cm, BC = 5 cm and AC = 6 cm.